

In [1]:

```
pip install numpy matplotlib scikit-learn scipy
```

```
Requirement already satisfied: numpy in
/usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in
/usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: scikit-learn in
/usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: scipy in
/usr/local/lib/python3.12/dist-packages (1.16.3)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib) (26.2)
Requirement already satisfied: pillow>=8 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib) (2.9.0.post0)
Requirement already satisfied: joblib>=1.2.0 in
/usr/local/lib/python3.12/
```

```
dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from
scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from
python-dateutil>=2.7->matplotlib) (1.17.0)
```

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram,
linkage
```

Dataset

```
X = np.array([
    [2, 10], # A1
    [2, 5], # A2
    [8, 4], # A3
    [5, 8], # B1
    [7, 5], # B2
    [6, 4], # B3
    [1, 2], # C1
    [4, 9] # C2
])
```

```
labels_names = ['A1', 'A2', 'A3', 'B1', 'B2', 'B3', 'C1',
                 'C2']
```

Agglomerative Clustering

```
hc = AgglomerativeClustering(
    n_clusters=3,
    linkage='ward'
)
```

```
cluster_labels = hc.fit_predict(X)
```

In [7]:

file:///home/mvsvr/Downloads/Hierarchical_(Agglomerative).html 1/3

06/06/2026, 13:34 Hierarchical_(Agglomerative)

Display Cluster Assignments

```
print("Cluster Assignments:")
```

```
for i in range(len(X)):
```

```
    print(f'{labels_names[i]} {tuple(X[i])} -> Cluster {cluster_labels[i]}
```

Scatter Plot

```
plt.figure(figsize=(8, 6))
```

```
plt.scatter(X[:, 0], X[:, 1], c=cluster_labels, cmap='rainbow', s=100)
```

```
for i, txt in enumerate(labels_names):
```

```
    plt.annotate(txt, (X[i, 0], X[i, 1]))
```

```
plt.title("Hierarchical Agglomerative Clustering")
plt.xlabel("X Coordinate")
plt.ylabel("Y Coordinate")
plt.grid(True)
plt.show()
```

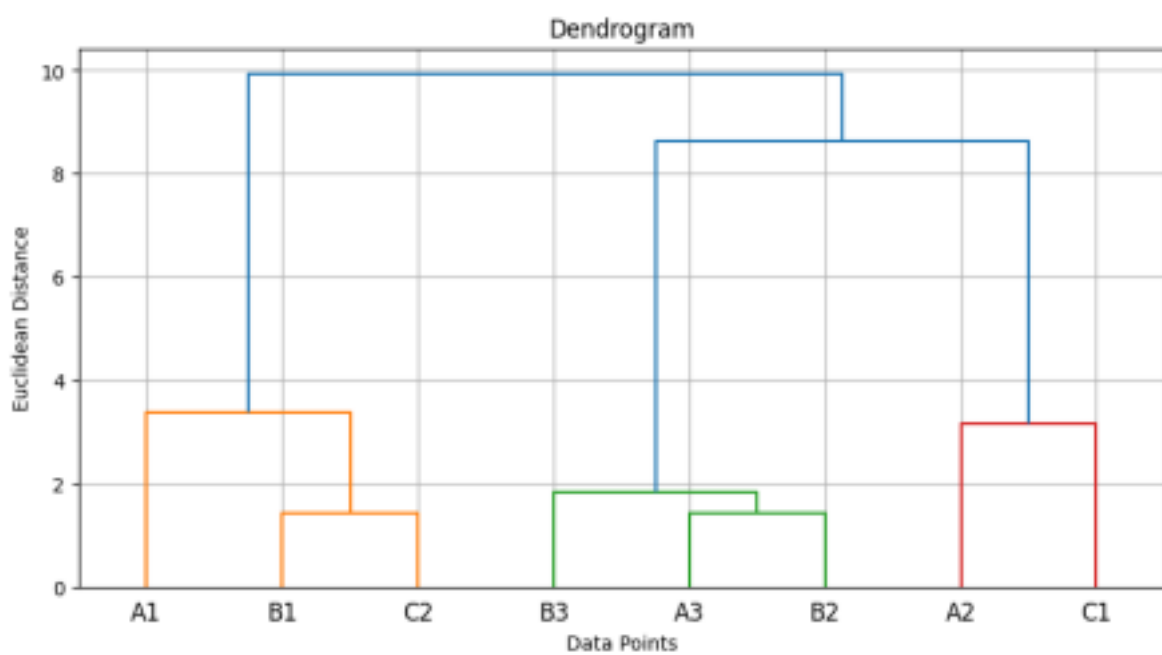
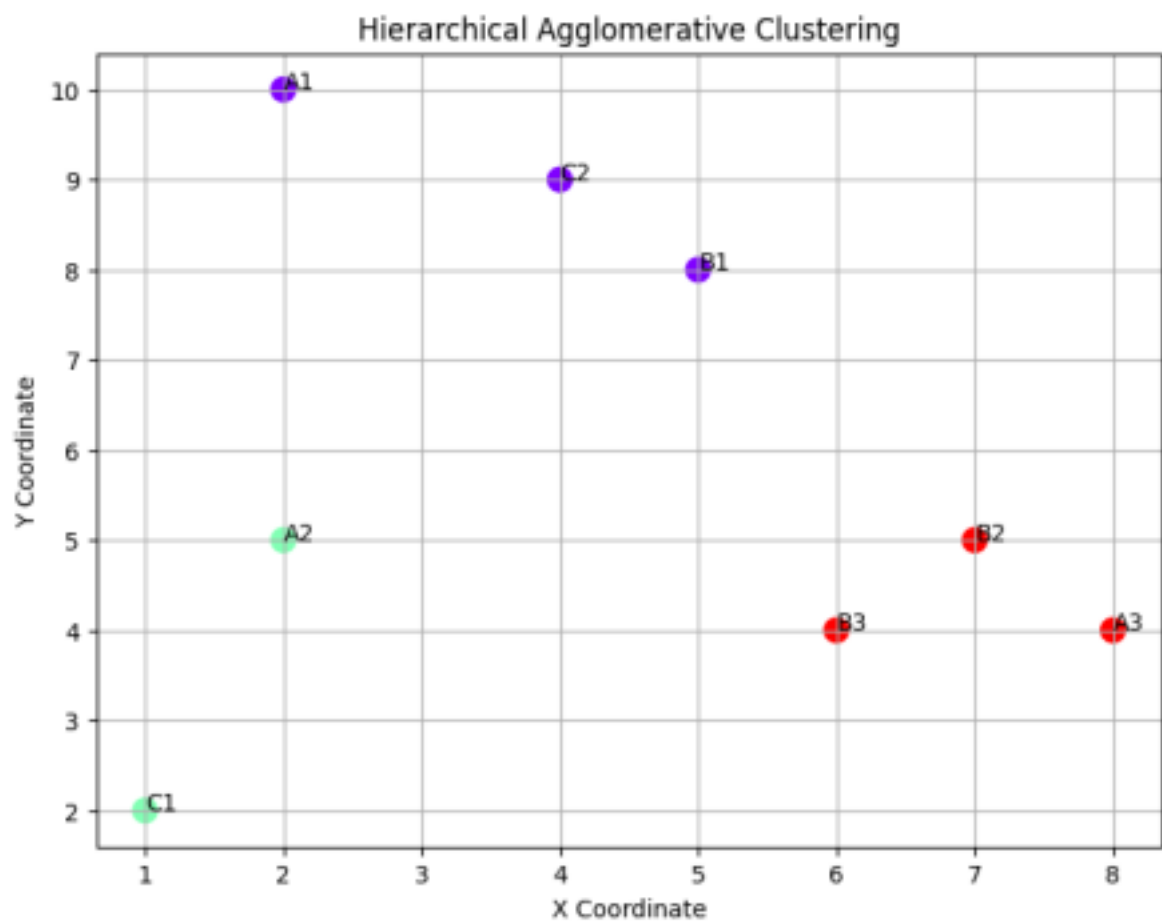
```
# Dendrogram
```

```
Z = linkage(X, method='ward')
```

```
plt.figure(figsize=(10, 5))
dendrogram(Z, labels=labels_names)
plt.title("Dendrogram")
plt.xlabel("Data Points")
plt.ylabel("Euclidean Distance")
plt.grid(True)
plt.show()
```

Cluster Assignments:

```
A1 (np.int64(2), np.int64(10)) -> Cluster 0
A2 (np.int64(2), np.int64(5)) -> Cluster 1
A3 (np.int64(8), np.int64(4)) -> Cluster 2
B1 (np.int64(5), np.int64(8)) -> Cluster 0
B2 (np.int64(7), np.int64(5)) -> Cluster 2
B3 (np.int64(6), np.int64(4)) -> Cluster 2
C1 (np.int64(1), np.int64(2)) -> Cluster 1
C2 (np.int64(4), np.int64(9)) -> Cluster 0
```



In []: